HR Attrition Analysis

Introduction

Employee attrition is one of the major challenges that organizations face because it can impact productivity, increase hiring costs, and affect team stability. Understanding the reasons behind employee turnover is important for improving employee satisfaction and retention.

In this project, we analyze the **IBM HR Analytics Employee Attrition dataset** to explore the key factors that influence employee attrition. The analysis focuses on identifying patterns and relationships between attrition and different employee characteristics such as age, income, overtime, and job role.

The project includes several steps such as **data exploration, data cleaning, exploratory data analysis (EDA), and machine learning modeling** to build a predictive model that can estimate the likelihood of employee attrition.

The insights generated from this analysis can help HR departments make data-driven decisions to improve employee retention and organizational performance.

Project Objectives

- Analyze employee attrition patterns
- Identify factors that influence employee turnover
- Explore relationships between attrition and employee attributes
- Build a machine learning model to predict employee attrition

Import Libraries

In [1]:
```
# Import libraries for data analysis and visualization
import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt
```

Load Dataset

In [2]:
```
# Load the dataset
df = pd.read_csv(r"D:\gergess_magdy\Data Analysis\projects_Data_analysis\3_Python_p
```

```
# Display the first 5 rows
df.head()
```

Out[2]:

	Age	Attrition	BusinessTravel	DailyRate	Department	DistanceFromHome	Education
0	41	Yes	Travel_Rarely	1102	Sales	1	2
1	49	No	Travel_Frequently	279	Research & Development	8	1
2	37	Yes	Travel_Rarely	1373	Research & Development	2	2
3	33	No	Travel_Frequently	1392	Research & Development	3	4
4	27	No	Travel_Rarely	591	Research & Development	2	1

5 rows × 35 columns

Data Overview

In [9]:

```
# Display dataset information (columns, data types, non-null counts)
df.info()

# Display statistical summary of numeric columns
df.describe()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1470 entries, 0 to 1469
Data columns (total 33 columns):
 #   Column                    Non-Null Count  Dtype
---  ------                    --------------  -----
 0   Age                       1470 non-null   int64
 1   Attrition                 1470 non-null   object
 2   BusinessTravel            1470 non-null   object
 3   DailyRate                 1470 non-null   int64
 4   Department                1470 non-null   object
 5   DistanceFromHome          1470 non-null   int64
 6   Education                 1470 non-null   int64
 7   EducationField            1470 non-null   object
 8   EmployeeNumber            1470 non-null   int64
 9   EnvironmentSatisfaction   1470 non-null   int64
 10  Gender                    1470 non-null   object
 11  HourlyRate                1470 non-null   int64
 12  JobInvolvement            1470 non-null   int64
 13  JobLevel                  1470 non-null   int64
 14  JobRole                   1470 non-null   object
 15  JobSatisfaction           1470 non-null   int64
 16  MaritalStatus             1470 non-null   object
 17  MonthlyIncome             1470 non-null   int64
 18  MonthlyRate               1470 non-null   int64
 19  NumCompaniesWorked        1470 non-null   int64
 20  Over18                    1470 non-null   object
 21  OverTime                  1470 non-null   object
 22  PercentSalaryHike         1470 non-null   int64
 23  PerformanceRating         1470 non-null   int64
 24  RelationshipSatisfaction  1470 non-null   int64
 25  StockOptionLevel          1470 non-null   int64
 26  TotalWorkingYears         1470 non-null   int64
 27  TrainingTimesLastYear     1470 non-null   int64
 28  WorkLifeBalance           1470 non-null   int64
 29  YearsAtCompany            1470 non-null   int64
 30  YearsInCurrentRole        1470 non-null   int64
 31  YearsSinceLastPromotion   1470 non-null   int64
 32  YearsWithCurrManager      1470 non-null   int64
dtypes: int64(24), object(9)
memory usage: 379.1+ KB
```

Out[9]:

	Age	DailyRate	DistanceFromHome	Education	EmployeeNumber	Envir
count	1470.000000	1470.000000	1470.000000	1470.000000	1470.000000	
mean	36.923810	802.485714	9.192517	2.912925	1024.865306	
std	9.135373	403.509100	8.106864	1.024165	602.024335	
min	18.000000	102.000000	1.000000	1.000000	1.000000	
25%	30.000000	465.000000	2.000000	2.000000	491.250000	
50%	36.000000	802.000000	7.000000	3.000000	1020.500000	
75%	43.000000	1157.000000	14.000000	4.000000	1555.750000	
max	60.000000	1499.000000	29.000000	5.000000	2068.000000	

8 rows × 24 columns

In [6]:
```
# Display the shape of the dataset (rows, columns)
df.shape
```

Out[6]: (1470, 35)

In [5]:
```
# Count of unique values per column
df.nunique()
```

Out[5]:
```
Age                          43
Attrition                     2
BusinessTravel                3
DailyRate                   886
Department                    3
DistanceFromHome             29
Education                     5
EducationField                6
EmployeeCount                 1
EmployeeNumber             1470
EnvironmentSatisfaction       4
Gender                        2
HourlyRate                   71
JobInvolvement                4
JobLevel                      5
JobRole                       9
JobSatisfaction               4
MaritalStatus                 3
MonthlyIncome              1349
MonthlyRate                1427
NumCompaniesWorked           10
Over18                        1
OverTime                      2
PercentSalaryHike            15
PerformanceRating             2
RelationshipSatisfaction      4
StandardHours                 1
StockOptionLevel              4
TotalWorkingYears            40
TrainingTimesLastYear         7
WorkLifeBalance               4
YearsAtCompany               37
YearsInCurrentRole           19
YearsSinceLastPromotion      16
YearsWithCurrManager         18
dtype: int64
```

In [7]:
```
# Check for missing values
df.isna().sum()
```

Out[7]:
```
Age                         0
Attrition                   0
BusinessTravel              0
DailyRate                   0
Department                  0
DistanceFromHome            0
Education                   0
EducationField              0
EmployeeCount               0
EmployeeNumber              0
EnvironmentSatisfaction     0
Gender                      0
HourlyRate                  0
JobInvolvement              0
JobLevel                    0
JobRole                     0
JobSatisfaction             0
MaritalStatus               0
MonthlyIncome               0
MonthlyRate                 0
NumCompaniesWorked          0
Over18                      0
OverTime                    0
PercentSalaryHike           0
PerformanceRating           0
RelationshipSatisfaction    0
StandardHours               0
StockOptionLevel            0
TotalWorkingYears           0
TrainingTimesLastYear       0
WorkLifeBalance             0
YearsAtCompany              0
YearsInCurrentRole          0
YearsSinceLastPromotion     0
YearsWithCurrManager        0
dtype: int64
```

Data Cleaning

In [3]:
```python
# Drop columns that are not useful for analysis
columns_to_drop = ['EmployeeCount', 'StandardHours', 'EmployeeNumber', 'Over18']
df = df.drop(columns=columns_to_drop, errors='ignore')

# Check the updated columns
print("Updated Columns:", df.columns.tolist())
```

```
Updated Columns: ['Age', 'Attrition', 'BusinessTravel', 'DailyRate', 'Department',
'DistanceFromHome', 'Education', 'EducationField', 'EnvironmentSatisfaction', 'Gende
r', 'HourlyRate', 'JobInvolvement', 'JobLevel', 'JobRole', 'JobSatisfaction', 'Marit
alStatus', 'MonthlyIncome', 'MonthlyRate', 'NumCompaniesWorked', 'OverTime', 'Percen
tSalaryHike', 'PerformanceRating', 'RelationshipSatisfaction', 'StockOptionLevel',
'TotalWorkingYears', 'TrainingTimesLastYear', 'WorkLifeBalance', 'YearsAtCompany',
'YearsInCurrentRole', 'YearsSinceLastPromotion', 'YearsWithCurrManager']
```

In [4]:
```
# Convert Attrition column from text to numeric
df['Attrition'] = df['Attrition'].map({'Yes': 1, 'No': 0})

# Verify encoding
df['Attrition'].value_counts()
```

Out[4]:
```
Attrition
0    1233
1     237
Name: count, dtype: int64
```

In [10]:
```
# Verify no missing values exist after cleaning
df.isnull().sum()
```

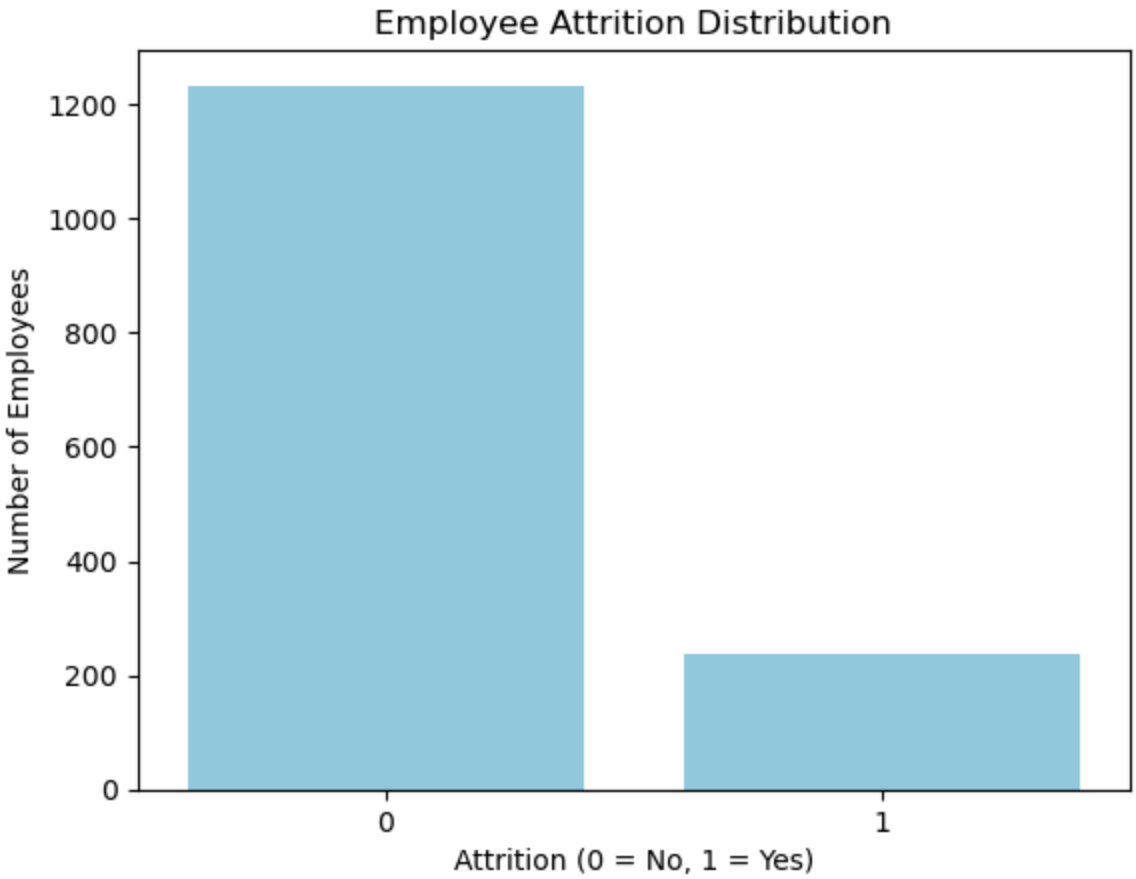
Out[10]:
```
Age                         0
Attrition                   0
BusinessTravel              0
DailyRate                   0
Department                  0
DistanceFromHome            0
Education                   0
EducationField              0
EnvironmentSatisfaction     0
Gender                      0
HourlyRate                  0
JobInvolvement              0
JobLevel                    0
JobRole                     0
JobSatisfaction             0
MaritalStatus               0
MonthlyIncome               0
MonthlyRate                 0
NumCompaniesWorked          0
OverTime                    0
PercentSalaryHike           0
PerformanceRating           0
RelationshipSatisfaction    0
StockOptionLevel            0
TotalWorkingYears           0
TrainingTimesLastYear       0
WorkLifeBalance             0
YearsAtCompany              0
YearsInCurrentRole          0
YearsSinceLastPromotion     0
YearsWithCurrManager        0
dtype: int64
```

df.dtypes

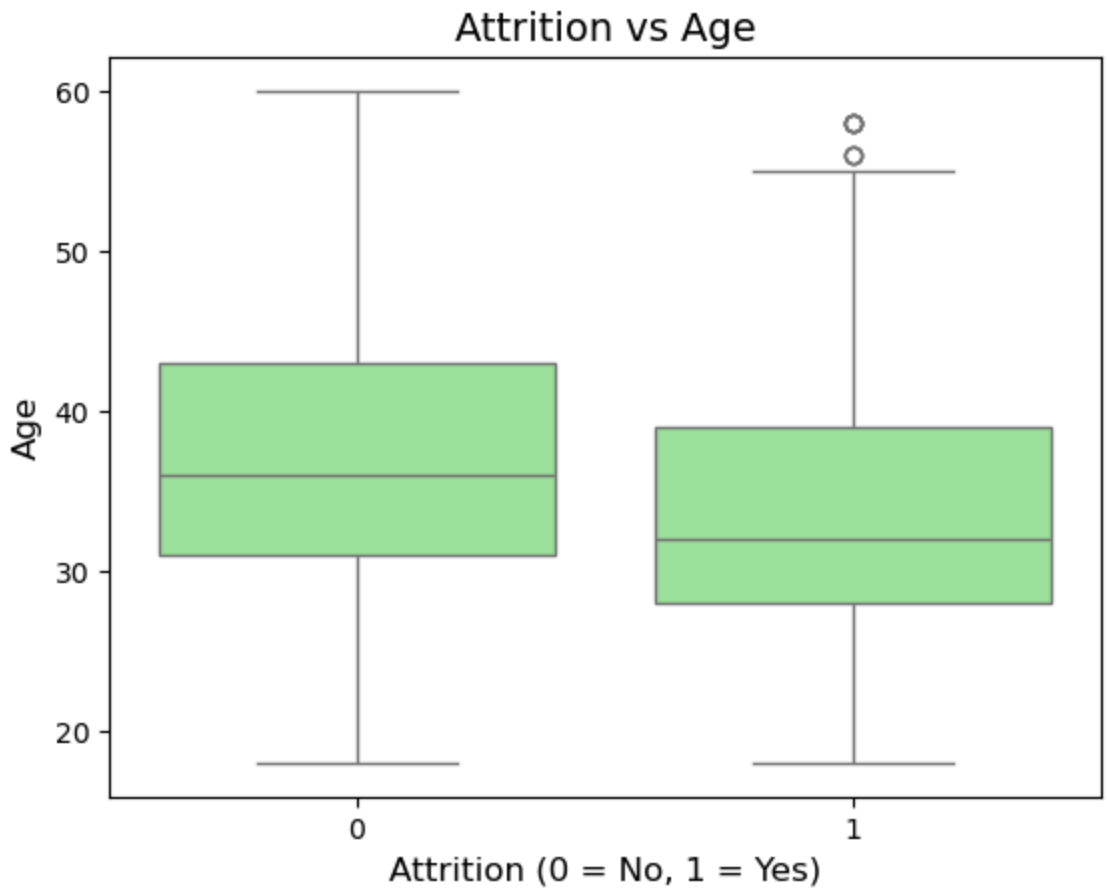
Exploratory Data Analysis (EDA)

In [5]:
```
# Plot Attrition Distribution
sns.countplot(data=df, x='Attrition', color='skyblue')
plt.title('Employee Attrition Distribution')
```

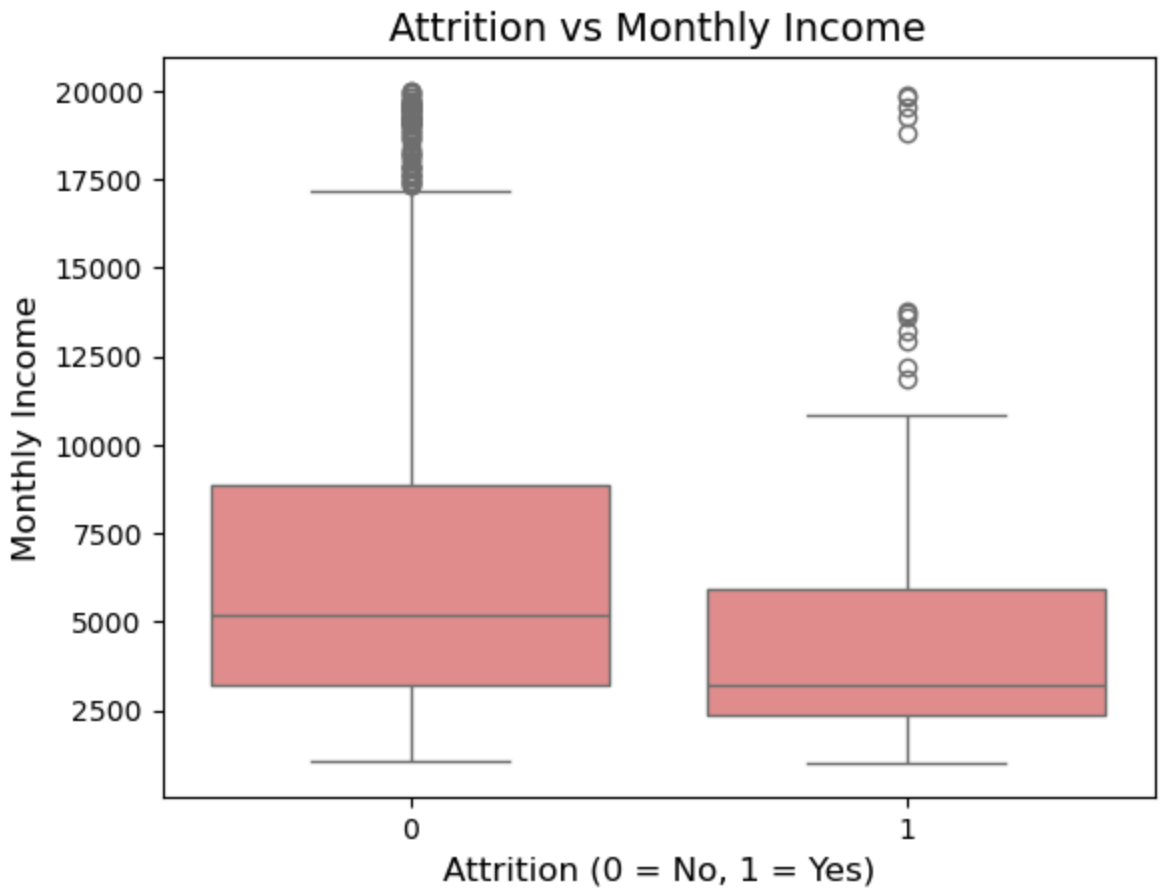
```
plt.xlabel('Attrition (0 = No, 1 = Yes)')
plt.ylabel('Number of Employees')
plt.show()
```



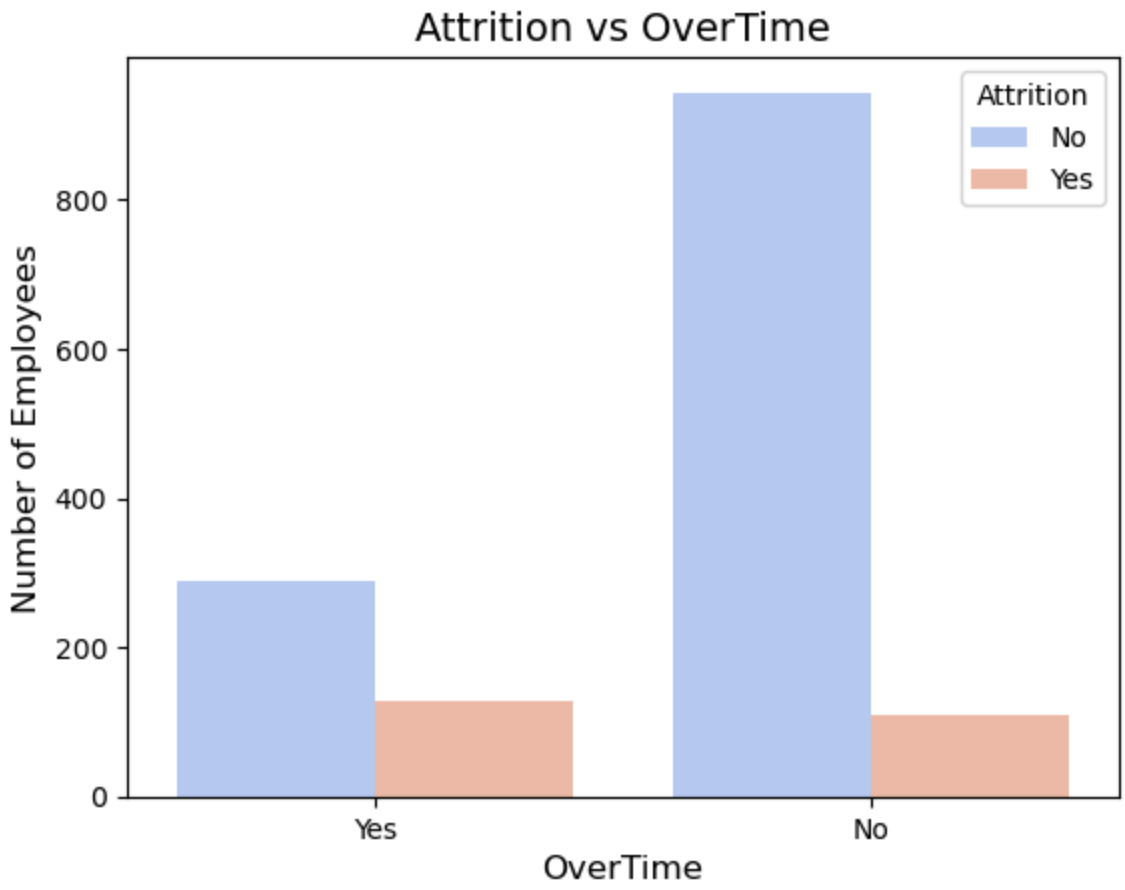
In [6]:
```
# Plot Attrition vs Age
sns.boxplot(x='Attrition', y='Age', data=df, color='lightgreen')
plt.title('Attrition vs Age', fontsize=14)
plt.xlabel('Attrition (0 = No, 1 = Yes)', fontsize=12)
plt.ylabel('Age', fontsize=12)
plt.show()
```



In [7]:
```
# Plot Attrition vs Monthly Income
sns.boxplot(x='Attrition', y='MonthlyIncome', data=df, color='lightcoral')
plt.title('Attrition vs Monthly Income', fontsize=14)
plt.xlabel('Attrition (0 = No, 1 = Yes)', fontsize=12)
plt.ylabel('Monthly Income', fontsize=12)
plt.show()
```



In [8]:
```
# Plot Attrition vs OverTime
sns.countplot(x='OverTime', hue='Attrition', data=df, palette='coolwarm')
plt.title('Attrition vs OverTime', fontsize=14)
plt.xlabel('OverTime', fontsize=12)
plt.ylabel('Number of Employees', fontsize=12)
plt.legend(title='Attrition', labels=['No','Yes'])
plt.show()
```



Correlation Analysis

To understand which features are most related to employee attrition, we calculated the correlation between all numerical variables and the target variable (Attrition).

The visualization below shows the top features that have the strongest relationship with employee attrition.

In [9]:
```
corr = df.corr(numeric_only=True)['Attrition'].sort_values(ascending=False)

corr
```

Out[9]:
```
Attrition                   1.000000
DistanceFromHome            0.077924
NumCompaniesWorked          0.043494
MonthlyRate                 0.015170
PerformanceRating           0.002889
HourlyRate                 -0.006846
PercentSalaryHike          -0.013478
Education                  -0.031373
YearsSinceLastPromotion    -0.033019
RelationshipSatisfaction   -0.045872
DailyRate                  -0.056652
TrainingTimesLastYear      -0.059478
WorkLifeBalance            -0.063939
EnvironmentSatisfaction    -0.103369
JobSatisfaction            -0.103481
JobInvolvement             -0.130016
YearsAtCompany             -0.134392
StockOptionLevel           -0.137145
YearsWithCurrManager       -0.156199
Age                        -0.159205
MonthlyIncome              -0.159840
YearsInCurrentRole         -0.160545
JobLevel                   -0.169105
TotalWorkingYears          -0.171063
Name: Attrition, dtype: float64
```

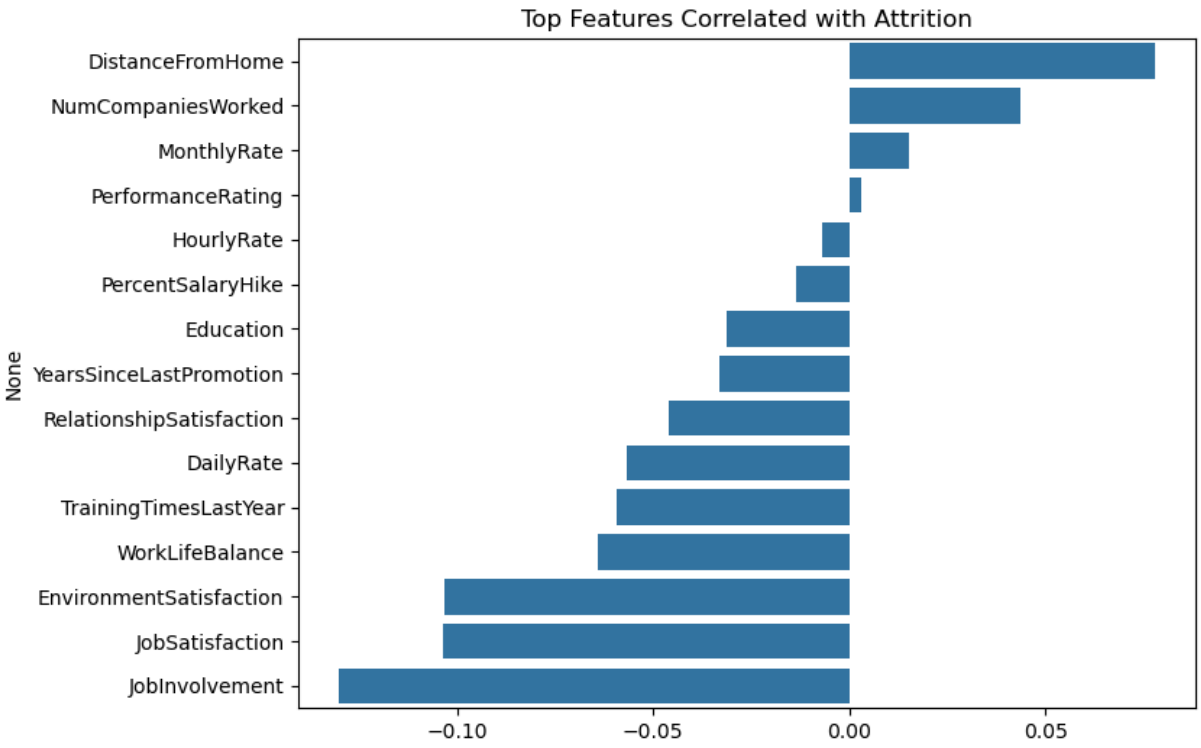
In [10]:
```
top_corr = corr[1:16]

plt.figure(figsize=(8,6))

sns.barplot(
    x=top_corr.values,
    y=top_corr.index
)

plt.title("Top Features Correlated with Attrition")

plt.show()
```



Key Insights from EDA

Based on the exploratory data analysis, here are the main observations regarding employee attrition:

1. **Attrition Rate is Low**
   Most employees stayed with the company, with a smaller proportion leaving (Attrition = 1).

2. **Younger Employees Tend to Leave More**
   Boxplots show that employees who leave are generally younger compared to those who stay.

3. **Lower Monthly Income is Associated with Higher Attrition**
   Employees with lower salaries are more likely to leave, indicating pay is a key factor.

4. **Overtime Increases Attrition Risk**
   Employees working overtime have a higher likelihood of leaving, suggesting workload affects retention.

5. **Attrition is Less Dependent on Age after 40**
   Employees above 40 years old show a lower turnover rate, indicating stability with experience.

6. **Work-Life Balance Factors May Influence Attrition**
   Although not directly visualized, the correlation with Overtime suggests that work-life

balance is important.

7. **Insights Can Inform HR Decisions**
   These patterns can help HR focus on retention strategies, salary reviews, and overtime management.

Predicting Employee Attrition with XGBoost

In this section, we build a machine learning model using **XGBoost** to predict which employees are likely to leave the company.

In [21]:
```
# Import libraries
from sklearn.model_selection import train_test_split
from sklearn.metrics import accuracy_score, confusion_matrix, classification_report
from imblearn.over_sampling import SMOTE
from xgboost import XGBClassifier

# Encode categorical variables
categorical_cols = df.select_dtypes(include=['object']).columns.tolist()
df_encoded = pd.get_dummies(df, columns=categorical_cols, drop_first=True)

# Separate features and target
X = df_encoded.drop('Attrition', axis=1)
y = df_encoded['Attrition']

# Split into train and test sets (stratified)
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_sta

# Apply SMOTE on training data
smote = SMOTE(random_state=42)
X_train_res, y_train_res = smote.fit_resample(X_train, y_train)

# Check class distribution
print("Before SMOTE:\n", y_train.value_counts())
print("After SMOTE:\n", y_train_res.value_counts())
```

```
Before SMOTE:
 Attrition
0    986
1    190
Name: count, dtype: int64
After SMOTE:
 Attrition
0    986
1    986
Name: count, dtype: int64
```

In [22]:
```
# Initialize and train XGBoost on balanced data
xgb_model = XGBClassifier(
    n_estimators=300,
    max_depth=6,
    learning_rate=0.1,
```

```
    random_state=42,
    eval_metric='logloss'
)
xgb_model.fit(X_train_res, y_train_res)
```

Out[22]:

```
▾ XGBClassifier
▶ Parameters
```

In [23]:

```
# Predict probabilities on test set
y_probs = xgb_model.predict_proba(X_test)[:, 1]

# Apply custom threshold
threshold = 0.3
y_pred = (y_probs >= threshold).astype(int)

# Evaluate results
print("Accuracy:", accuracy_score(y_test, y_pred))
print("Confusion Matrix:\n", confusion_matrix(y_test, y_pred))
print("Classification Report:\n", classification_report(y_test, y_pred))
```

```
Accuracy: 0.8401360544217688
Confusion Matrix:
 [[227  20]
 [ 27  20]]
Classification Report:
               precision    recall  f1-score   support

           0       0.89      0.92      0.91       247
           1       0.50      0.43      0.46        47

    accuracy                           0.84       294
   macro avg       0.70      0.67      0.68       294
weighted avg       0.83      0.84      0.83       294
```

Feature Importance Analysis

In [24]:

```
# Create a dataframe for feature importance
feature_importance = pd.DataFrame({
    'Feature': X_train.columns,
    'Importance': xgb_model.feature_importances_
})

# Sort features by importance
feature_importance = feature_importance.sort_values(by='Importance', ascending=Fals

# Show top 10 most important features
feature_importance.head(10)
```

Out[24]:

	Feature	Importance
15	StockOptionLevel	0.123562
41	MaritalStatus_Married	0.072798
26	Department_Sales	0.070440
7	JobLevel	0.062096
42	MaritalStatus_Single	0.061697
33	JobRole_Human Resources	0.044742
8	JobSatisfaction	0.042804
25	Department_Research & Development	0.040448
31	EducationField_Technical Degree	0.032943
29	EducationField_Medical	0.027608

In [25]:

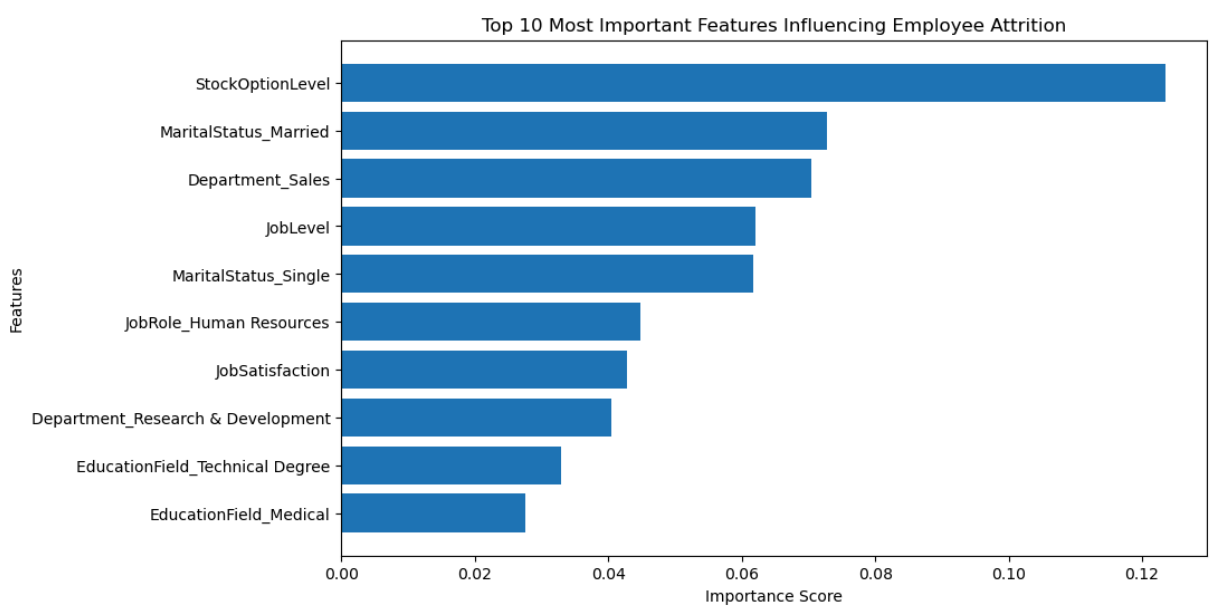
```
# Plot top 10 important features
plt.figure(figsize=(10,6))

plt.barh(
    feature_importance['Feature'].head(10),
    feature_importance['Importance'].head(10)
)

plt.gca().invert_yaxis()

plt.title("Top 10 Most Important Features Influencing Employee Attrition")
plt.xlabel("Importance Score")
plt.ylabel("Features")

plt.show()
```



Key Drivers of Employee Attrition

Based on the XGBoost feature importance analysis, several factors appear to play a significant role in predicting employee attrition.

1. Stock Option Level

This is the most influential feature in the model. Employees with lower stock option benefits may be more likely to leave the company, suggesting that long-term financial incentives can improve retention.

2. Marital Status

Both "Married" and "Single" categories appear among the most important features, indicating that personal life circumstances may influence job stability and turnover behavior.

3. Department

Employees in the **Sales** and **Research & Development** departments show notable influence on attrition predictions. This may suggest different workload levels or career pressures across departments.

4. Job Level

Job level is an important factor, which may indicate that employees in certain seniority levels are more likely to leave due to career growth opportunities or job expectations.

5. Job Satisfaction

Lower job satisfaction is strongly associated with higher attrition risk, highlighting the importance of employee engagement and workplace satisfaction.

6. Education Field

Certain education backgrounds such as **Technical Degree** and **Medical fields** appear to influence attrition predictions, possibly reflecting varying job market demand for these skills.

Business Recommendations

Based on the analysis and machine learning results, several strategies can help reduce employee attrition:

1. Improve financial incentives such as stock options and compensation packages.

2. Monitor workload and overtime, especially in high-pressure departments like Sales.
3. Enhance job satisfaction through employee engagement programs.
4. Focus on retention strategies for employees in early career stages.
5. Provide career development opportunities and training programs.

Implementing these strategies can help organizations improve employee retention and reduce turnover.